Lab 2: Software Requirements Specification for Gold Star

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CS411W

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1 Introduction

1.1 Purpose

This document provides detailed specifications and requirements for Gold Star, a web-based habit-building and time management platform.

1.2 Scope

Millions of people struggle with ADHD symptoms such as starting tasks, staying focused, or finishing what they have started (CDC, 2024). They need a solution that is built with them in mind, allowing them the grace and flexibility needed to be successful. Gold Star introduces the scheduler to flexible scheduling, giving them the ability to change their routine when they need it, or are not feeling the most productive.

1.3 Definitions, Acronyms, and Abbreviations

Activity Alternation – Switching from an activity requiring high mental focus to an activity requiring low mental focus.

Activity Switching – The cognitive process of disengaging from one task and shifting mental focus to another task.

Adaptive Scheduling – To schedule a task which can be completed anytime within a set timeframe. For example, a task which can be completed anywhere inclusively between 7:00 P.M. and 10:00 P.M.

Application Programming Interface (API) - A set of standard rules, protocols, and definitions

allowing different software applications to interact with one another.

Attention-Deficit/Hyperactivity Disorder (ADHD) - A common neurodevelopmental disorder that alters the brain's ability to plan, focus, remember instructions, and manage emotions.

Cascading Style Sheets (CSS) - A style sheet language used to determine the visual presentation, layout, and aesthetics of a website.

Git - A digital version control system intended to preserve a history of saved modifications, additions, and deletions to software source code.

GitHub - A cloud-based platform where developers can collaborate on software code.

Gold Star - A web-based habit-building and time management platform designed specifically for individuals with ADHD.

Habit Stacking – Interspersing new habits with existing habits to make task completion easier without feeling overwhelmed.

Hyperactivity - A state of being unusually or abnormally active.

HyperText Markup Language (HTML) - The standard markup language used to create and structure webpages.

JavaScript (JS) - A lightweight programming language used to build dynamic interactivity on web pages.

Jest - An open-source JavaScript testing framework.

JSDoc - An API documentation generator and syntax standard for JavaScript.

Pomodoro Technique – A time-management technique which separates work into 25-minute increments and 5-minute breaks. The standard Pomodoro process repeats this technique 4 times, then a longer break between 20 to 30 minutes is taken before the process repeats.

PostgreSQL - An open-source RDBMS that stores, manages, and retrieves data.

Reinforce - To make something stronger, more effective, or more compelling by adding support.

Relational Database Management System (RBMS) - Software used to store, manage, query, and retrieve data structured in table form.

Rigid Scheduling – To schedule a high-priority task which must either be initiated at or completed before an established time. For example, a task which must start at 6:00 P.M. or must be completed by a deadline of 6:00 P.M.

Routine Adaption – A requirement for an individual to change one's schedule or create new routines.

Routine - A sequence of actions regularly followed, or a fixed program.

Task Deviation – When a task has an unexpected change regarding completion. A task may be accomplished out of order in one's schedule, or it may not be accomplished at all.

Visual Studio Code (VSCode) - A development environment used for software engineering and web development.

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1.5 Overview

Section 2 of this document provides a brief overview of Gold Star.

Section 3 contains specific feature requirements.

2 Overall Description

2.1 Product Perspective

Gold Star is a web-based time management platform that allows those who struggle with ADHD symptoms to manage their time, tasks, and daily responsibilities. With scheduling that incorporates researched productivity boosting techniques and habit-building strategies, Gold Star helps schedulers stay consistent, complete tasks, and better manage their daily lives.

2.2 Product Functions

The Gold Star main user interface consists of a calendar view that allows the scheduler to clearly see their curated schedule with the ability to add, edit, or remove tasks from it. This interface also allows for task prioritization and grouping of tasks. This curated schedule is created with Gold Star's unique Adaptive Scheduling system which allows changes to priorities and rescheduling missed tasks when needed via Gold Star's "Roll Over" system. This allows the

scheduler to move their missed tasks to another day, giving them the flexibility they need without any added stress.

2.3 User Classes and Characteristics

Gold Star has one main class of users: schedulers. Schedulers add, edit, and remove tasks from the calendar view interface, choose their productivity level, and manipulate their schedule when needed.

2.4 Design and Implementation Constraints

N/A

2.5 Assumptions and Dependencies

N/A